

Non-Gaussian Response Distributions

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Overview

GAMs are not limited to Gaussian responses. By specifying a **family** (distribution) and **link function**, we can model count data, binary outcomes, positive continuous data, and more. The general model is:

$$g(\mu_i) = \beta_0 + f_1(x_{1i}) + \cdots + f_p(x_{pi}), \quad y_i \sim \text{Family}(\mu_i, \phi)$$

This vignette demonstrates three common non-Gaussian families:

Family	Link	Response type	Example
Poisson	log	Counts	Species counts, event rates
Binomial	logit	Binary / proportions	Presence-absence, disease status
Gamma	inverse (or log)	Positive continuous	Waiting times, claim sizes

Setup

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

```
library(gratia)
library(ggplot2)
```

Poisson GAM — count data

Loading count data

We load counts generated from a Poisson distribution with a smooth log-rate:

$$y_i \sim \text{Poisson}(\mu_i), \quad \log(\mu_i) = 1 + 1.5 \sin(2\pi x_i)$$

```
dat_pois <- read.csv("../data_poisson.csv")
x <- dat_pois$x
y_pois <- dat_pois$y
n <- nrow(dat_pois)
eta <- 1.0 + 1.5 * sin(2 * pi * x)

df_pois <- data.frame(x = x, y = y_pois)
```

Fitting the model

```
m_pois <- gam(y ~ s(x, k = 15, bs = "cr"), data = df_pois,  
             family = poisson(link = "log"), method = "REML")  
summary(m_pois)
```

Family: poisson

Link function: log

Formula:

y ~ s(x, k = 15, bs = "cr")

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.93959	0.04685	20.06	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(x)	7.995	9.613	719.9	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

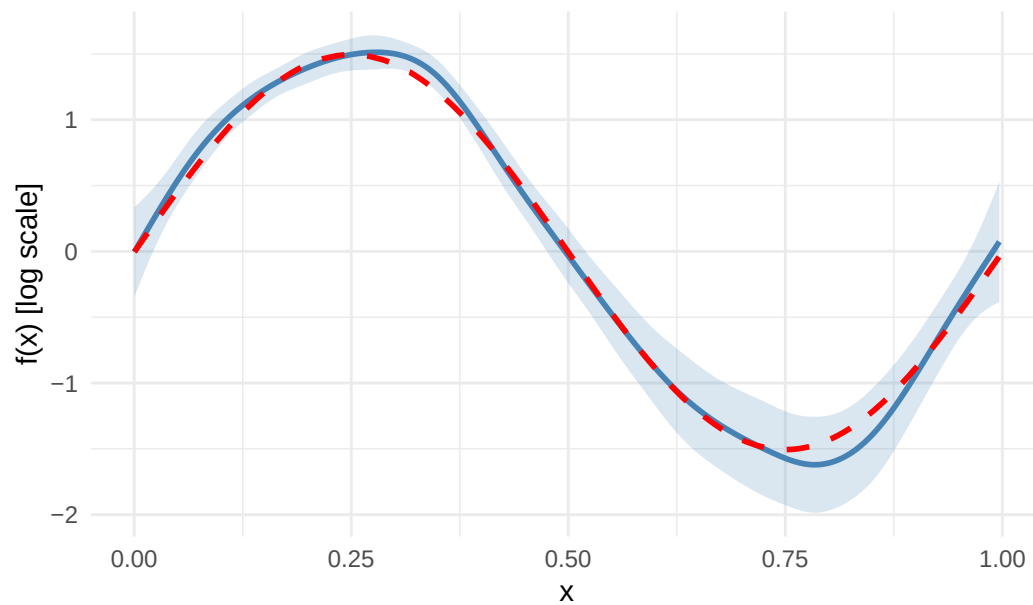
R-sq.(adj) = 0.764 Deviance explained = 76.4%

-REML = 566.91 Scale est. = 1 n = 300

Smooth estimate

```
se_pois <- smooth_estimates(m_pois, n = 200)  
  
ggplot(se_pois, aes(x = x, y = .estimate)) +  
  geom_ribbon(aes(ymin = .estimate - 2 * .se, ymax = .estimate + 2 * .se),  
            alpha = 0.2, fill = "steelblue") +  
  geom_line(linewidth = 1, colour = "steelblue") +  
  geom_line(data = data.frame(x = x, y = eta - mean(eta)),  
            aes(x = x, y = y), linetype = "dashed", linewidth = 1, colour = "red") +  
  labs(x = "x", y = "f(x) [log scale]", title = "Poisson GAM - smooth on link scale") +  
  theme_minimal()
```

Poisson GAM – smooth on link scale

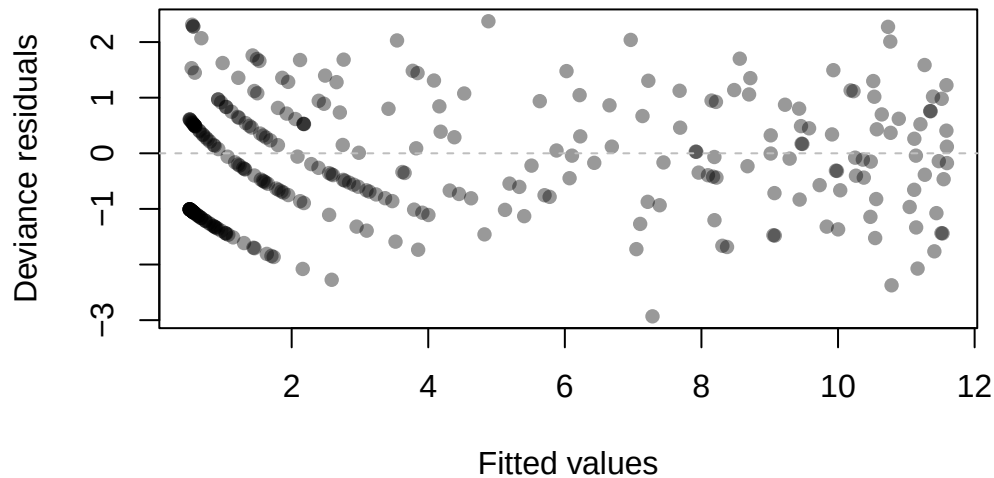


Deviance residuals

```
resid_pois <- residuals(m_pois, type = "deviance")

plot(fitted(m_pois), resid_pois,
     xlab = "Fitted values", ylab = "Deviance residuals",
     main = "Poisson GAM - residuals", pch = 16, col = adjustcolor("black", 0.4))
abline(h = 0, lty = 2, col = "grey")
```

Poisson GAM – residuals



Binomial GAM — binary data

Loading binary data

We load binary outcomes generated from a logistic model:

$$y_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = -0.5 + 2 \sin(2\pi x_i)$$

```
dat_binom <- read.csv("../data_binomial.csv")
x_bin <- dat_binom$x
y_bin <- dat_binom$y
eta_bin <- -0.5 + 2.0 * sin(2 * pi * x_bin)
df_bin <- data.frame(x = x_bin, y = y_bin)
```

Fitting the model

```
m_bin <- gam(y ~ s(x, k = 15, bs = "cr"), data = df_bin,
             family = binomial(link = "logit"), method = "REML")
summary(m_bin)
```

Family: binomial
Link function: logit

Formula:
 $y \sim s(x, k = 15, bs = "cr")$

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.8588	0.1599	-5.371	7.83e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(x)	5.034	6.217	69.7	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

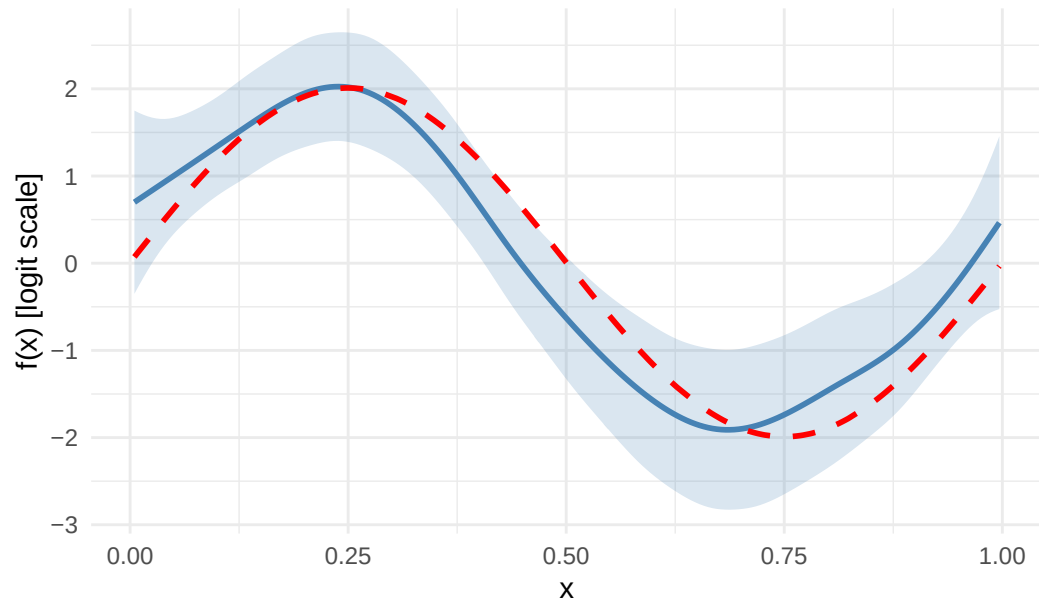
R-sq.(adj) = 0.295 Deviance explained = 25.5%
-REML = 153.02 Scale est. = 1 n = 300

Smooth estimate

```
se_bin <- smooth_estimates(m_bin, n = 200)

ggplot(se_bin, aes(x = x, y = .estimate)) +
  geom_ribbon(aes(ymin = .estimate - 2 * .se, ymax = .estimate + 2 * .se),
            alpha = 0.2, fill = "steelblue") +
  geom_line(linewidth = 1, colour = "steelblue") +
  geom_line(data = data.frame(x = x_bin, y = eta_bin - mean(eta_bin)),
            aes(x = x, y = y), linetype = "dashed", linewidth = 1, colour = "red") +
  labs(x = "x", y = "f(x) [logit scale]", title = "Binomial GAM - smooth on link scale") +
  theme_minimal()
```

Binomial GAM – smooth on link scale

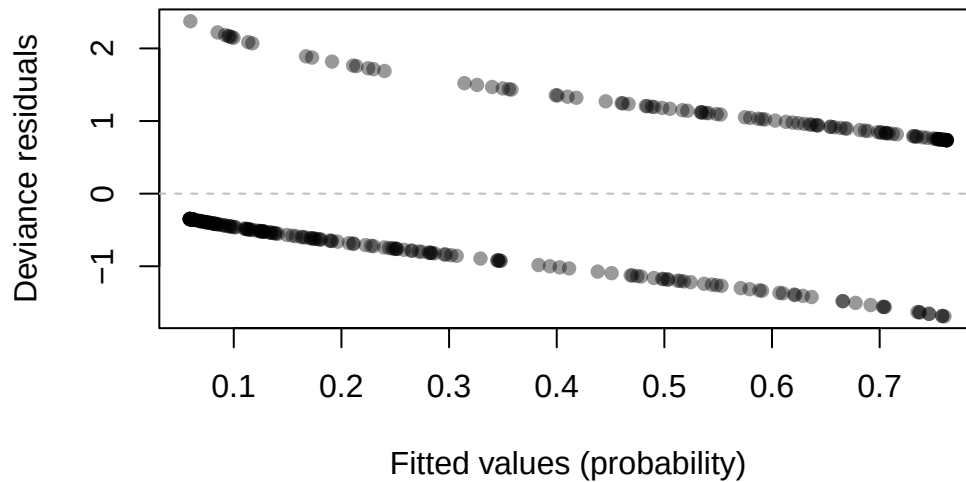


Deviance residuals

```
resid_bin <- residuals(m_bin, type = "deviance")

plot(fitted(m_bin), resid_bin,
     xlab = "Fitted values (probability)", ylab = "Deviance residuals",
     main = "Binomial GAM - residuals", pch = 16, col = adjustcolor("black", 0.4))
abline(h = 0, lty = 2, col = "grey")
```

Binomial GAM – residuals



Gamma GAM — positive continuous data

Loading positive continuous data

We load positive continuous data generated from a Gamma distribution with a smooth log-mean:

$$y_i \sim \text{Gamma}(\text{shape}, \text{scale}_i), \quad \log(\mu_i) = 1 + \sin(2\pi x_i)$$

```
dat_gamma <- read.csv("../data_gamma.csv")
x_gamma <- dat_gamma$x
y_gamma <- dat_gamma$y
eta_gamma <- 1.0 + sin(2 * pi * x_gamma)

df_gamma <- data.frame(x = x_gamma, y = y_gamma)
```

Fitting the model

We use Gamma with log link (more commonly used in practice than the canonical inverse link):

```
m_gamma <- gam(y ~ s(x, k = 15, bs = "cr"), data = df_gamma,
               family = Gamma(link = "log"), method = "REML")
summary(m_gamma)
```


Family: Gamma
Link function: log

Formula:
 $y \sim s(x, k = 15, bs = "cr")$

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.02615	0.02457	41.76	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(x)	6.177	7.594	30.27	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

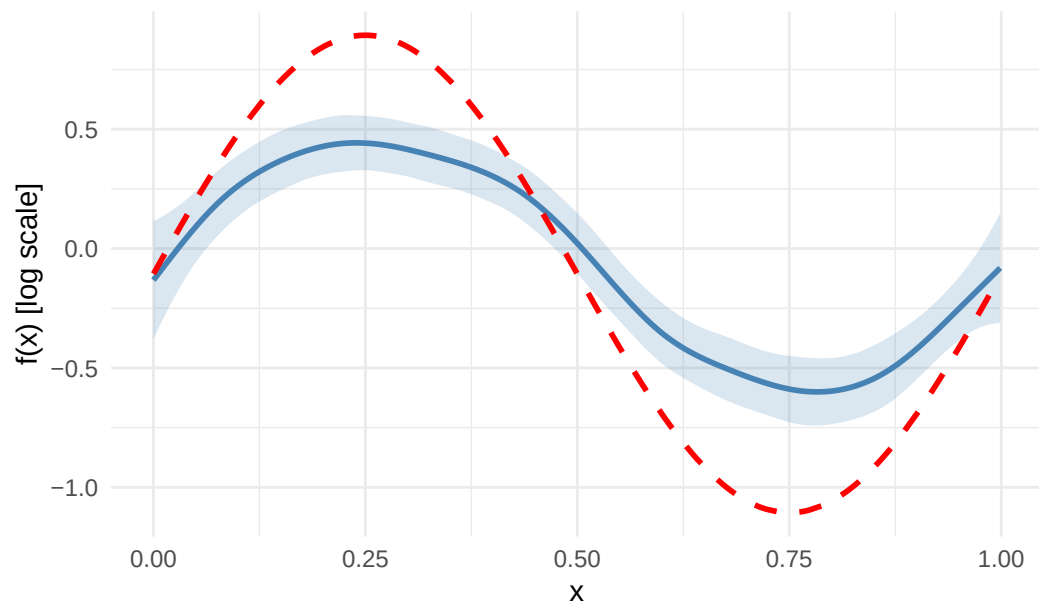
R-sq.(adj) = 0.362 Deviance explained = 41.5%
-REML = 473.96 Scale est. = 0.18114 n = 300

Smooth estimate

```
se_gam <- smooth_estimates(m_gam, n = 200)

ggplot(se_gam, aes(x = x, y = .estimate)) +
  geom_ribbon(aes(ymin = .estimate - 2 * .se, ymax = .estimate + 2 * .se),
            alpha = 0.2, fill = "steelblue") +
  geom_line(linewidth = 1, colour = "steelblue") +
  geom_line(data = data.frame(x = x_gam, y = eta_gam - mean(eta_gam)),
            aes(x = x, y = y), linetype = "dashed", linewidth = 1, colour = "red") +
  labs(x = "x", y = "f(x) [log scale]", title = "Gamma GAM - smooth on link scale") +
  theme_minimal()
```

Gamma GAM – smooth on link scale

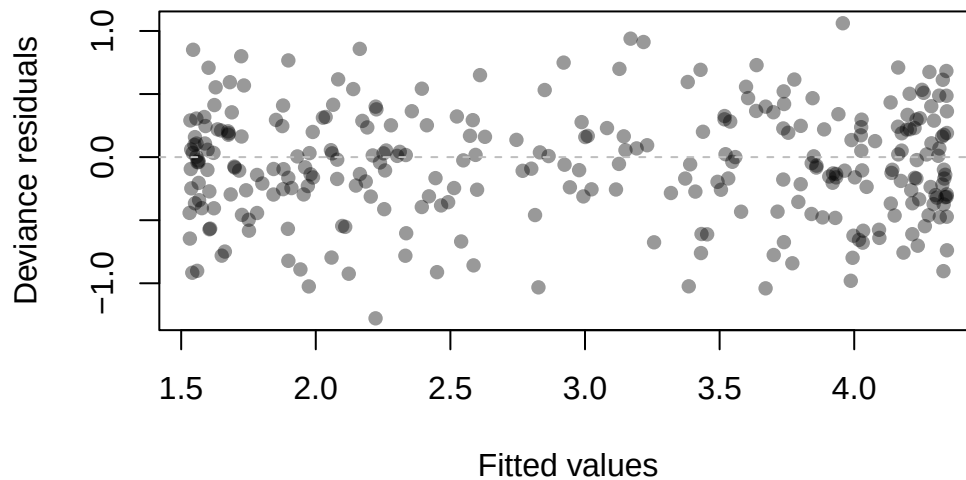


Deviance residuals

```
resid_gam <- residuals(m_gam, type = "deviance")

plot(fitted(m_gam), resid_gam,
     xlab = "Fitted values", ylab = "Deviance residuals",
     main = "Gamma GAM - residuals", pch = 16, col = adjustcolor("black", 0.4))
abline(h = 0, lty = 2, col = "grey")
```

Gamma GAM – residuals



Comparison

```
results <- data.frame(  
  Family = c("Poisson", "Binomial", "Gamma"),  
  EDF = c(  
    round(sum(pen.edf(m_pois)), 2),  
    round(sum(pen.edf(m_bin)), 2),  
    round(sum(pen.edf(m_gam)), 2)  
  ),  
  Dev_Explained = c(  
    round(summary(m_pois)$dev.expl * 100, 1),  
    round(summary(m_bin)$dev.expl * 100, 1),  
    round(summary(m_gam)$dev.expl * 100, 1)  
  ),  
  Scale = c(  
    round(summary(m_pois)$scale, 4),  
    round(summary(m_bin)$scale, 4),  
    round(summary(m_gam)$scale, 4)  
  )  
)  
print(results)
```

	Family	EDF	Dev_Explained	Scale
1	Poisson	8.00	76.4	1.0000

2	Binomial	5.03	25.5	1.0000
3	Gamma	6.18	41.5	0.1811

Summary

In this vignette we:

1. Simulated count data and fitted a **Poisson GAM** with a log link
2. Simulated binary data and fitted a **Binomial GAM** with a logit link
3. Simulated positive continuous data and fitted a **Gamma GAM** with a log link
4. Examined smooth estimates and deviance residuals for each family
5. Compared EDF and deviance explained across families

Each family uses a different link function to map the linear predictor to the mean of the response distribution, but the smooth estimation machinery is the same.